INDIAN INSTITUTE OF SCIENCE EDUCATION AND RESEARCH, MOHALI

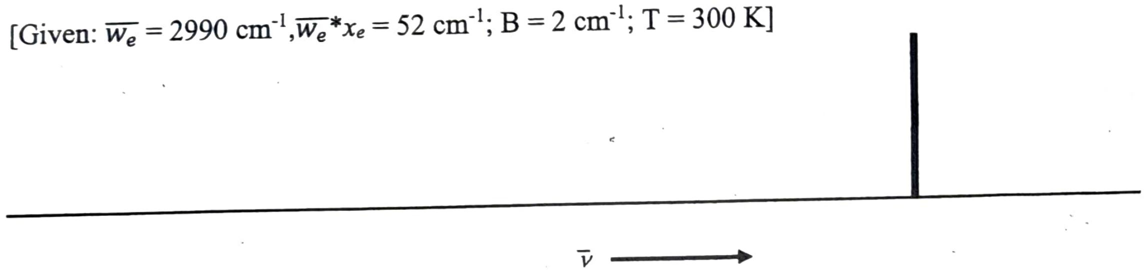
2nd Mid-SEMESTER 2025-2026

CHM201: SPECTROSCOPIC AND OTHERR PHYSICAL METHODS

FULL MARKS 15 **DURATION 1 HR**

1. The monochromatic laser used for Raman scattering of N_2O is marked below at 22000 cm^{-1}. Draw the peaks (as lines) and assign them as one can expect for pure rotational-Raman transitions and pure-vibrational Raman transition (Stokes Shift only) for the fundamental band due to symmetric stretch of the molecule. Please write clearly the values of energy-gaps along with the rotational energy states (*J*-values) for pure rotational-Raman and the energy gap for vibrational Raman shift for fundamental band. You can assume the molecule as rigid-rotor and ignore the correction for centrifugal distortion. Please follow the reference spectrum given below for your sketch. **4+ 2 = 6**

[Given: $\overline{w_e} = 2990$ cm^{-1}, $\overline{w_e}*x_e = 52$ cm^{-1}; B = 2 cm^{-1}; T = 300 K]



2. Which of the following molecules exhibit Rotational Raman and vibrational Raman spectra. Specify the vibrational bands for each which will be Raman inactive (if any).
(a) SF_6 ;(b) NH_3; (c) CH_4; (d) CH_3F; (e) O_3 **2.5 + 2.5 = 5**

3. An experimental Rotation-Vibration spectrum of $^1H\text{-}^{35}Cl$ for $v = 0$ to $v = 1$ transition is shown in Figure 1 and the corresponding band positions are given below: **4**

$P_0 = 8.60*10^{13}$ Hz ; $R_0 = 8.70*10^{13}$ Hz

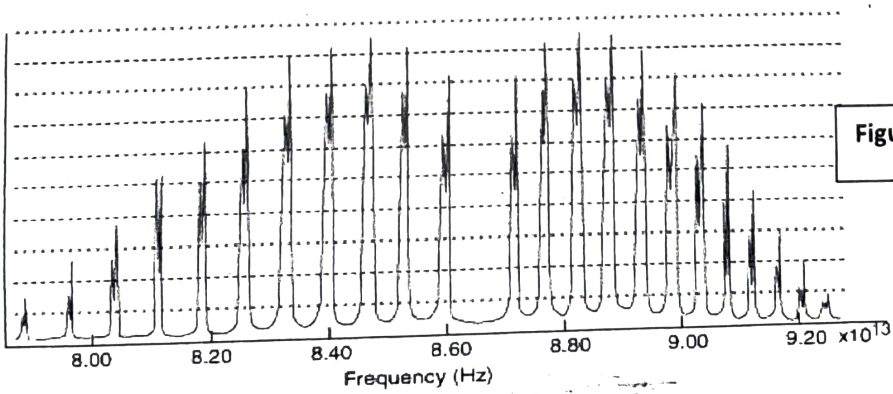


Figure 1

(a). Calculate the (i) rotational constant (in cm^{-1}), (ii) anharmonicity constant of $^1H\text{-}^{35}Cl$ from the spectrum. [Given the equilibrium oscillation frequency, $\overline{w_e}$ =2990 cm^{-1}]